

Towards mechanism-focused and scalable resilience interventions

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Abstract

Resilience interventions promote mental health by preparing individuals for adversity or building resilient outcomes during or after stress exposure. In this Review, we describe how currently available resilience interventions, focused on training individual psychosocial factors and lacking real-world implementation evidence, yield only small-to-medium favourable effects. Our synthesis of the literature supports a shift in resilience interventions research from training individual factors to training resilience mechanisms, in line with current theories of resilience. The effectiveness and reach of resilience interventions could be enhanced by targeting the mental, physiological and behavioural processes that occur during stressor exposure and by identifying the conditions under which different individuals benefit. We discuss an integrated approach for developing, testing and adapting resilience interventions in real-world settings.

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Introduction

Globally representative data from 146 countries¹ shows that the number of people reporting exposure to stress has doubled between 2006 and 2023. Stressors range from distressing events in everyday life (daily hassles) to potentially traumatic events (such as exposure to threatened death, serious injury or sexual violence²) and population-level stressors^{3,4}. These stressors span financial instability, pandemics, armed conflict and climate-driven weather extremes – frequently framed as manifestations of a global ‘polycrisis’ – that interact with geopolitical tensions to increase inequality and accelerate biodiversity loss⁵. About 70% of the population worldwide will experience at least one potentially traumatic event during their lifetime⁶, with an estimated 56–62% experiencing such an event before the age of 18 years^{7,8}. Because intense or prolonged stressors that overwhelm individuals’ coping resources⁹ play a central part in the onset and persistence of mental health conditions, increases in population-level stressor exposure challenge public mental health^{10–14}.

Nevertheless, approximately 66% of individuals show a resilient outcome following stressor exposure^{15–17}. Resilient outcomes are not defined solely by the absence of mental health conditions, such as post-traumatic stress disorder, during or after exposure to a stressor; they also encompass low levels of psychopathological symptoms (such as depressive or anxiety symptoms) and the presence of mental wellbeing and high levels of psychosocial functioning¹⁸. According to prevailing models in resilience research, resilient outcomes are partially facilitated by resilience factors, which are relatively stable psychosocial characteristics or resources at the individual, social (interpersonal relationships or daily interaction) or societal (systemic or structural aspects) levels^{19–22} that can also be assessed in the absence of stressor exposure. The influence of resilience factors on resilient outcomes is hypothesized to be mediated by a smaller set of resilience mechanisms, defined as the mental, physiological or behavioural processes activated during or following stressor exposure that increase an individual’s capacity to maintain or regain mental health during and/or after such exposure^{19,23,24}. Resilience factors would increase the probability that resilience mechanisms are activated, which in turn foster resilient outcomes^{19,20,25} (Fig. 1).

Resilience interventions are programmes, techniques, strategies and measures that seek to enhance resilient outcomes^{26–29}. Such interventions might be delivered before stressor exposure (primary preventive measures) as well as during or after exposure, when unfavourable outcomes might already be emerging^{26,30}. Therefore, resilience interventions hold promise for protecting individuals – and, at a broader scale, social groups and societies – from the adverse effects of increasing exposure to stressors³⁰ and might reduce the substantial personal and societal costs associated with psychological distress and mental health conditions^{31,32}.

However, current research on resilience interventions does not reflect prominent theoretical work that emphasizes the importance of resilience mechanisms over resilience factors and the value of multisystemic approaches to both factors and mechanisms. Dynamic developmental processes that influence resilient outcomes operate at multiple levels, including the individual, family, educational system, broader community and society³³. Similarly, research on resilience interventions must examine and build on the multiple processes that operate across these different systems, as well as their dynamic interactions.

In this Review, we build on state-of-the-art resilience theories and models to advance the development of interventions focused on mechanisms. First, we describe the evidence on the efficacy and effectiveness

of available interventions and discuss their limitations. Then, we critically examine the rationales that underlie these interventions. Next, we detail how the field of resilience intervention research can benefit from a shift from targeting individual resilience factors towards targeting the processes underlying resilient outcomes during stressor exposure. Finally, we outline the importance of variables that influence what works, when, under which conditions, and for whom to set the basis for the development of tailored interventions. We conclude by discussing recommendations towards an integrated development and evaluation of interventions in real-world settings that could substantially enhance the public health impact of resilience interventions.

Throughout the Review, we consider both consistently low levels of mental health symptoms and consistently high levels of psychosocial functioning and mental wellbeing as indicators of resilient outcomes¹⁸; that is, we do not adopt a binary approach to resilience by focusing on the presence or absence of mental health conditions or developing them belatedly. However, the Review covers evidence about symptoms rather than positive mental health outcomes; this evidence reflects an imbalance in the available literature, not a statement about the relative importance of these outcomes. Similarly, contextual and environmental resilience factors are not covered here because they are typically not addressed in resilience interventions. We focus on psychosocial resilience factors, although we acknowledge research on neuro-biological resilience factors^{19,34}, which are not the primary focus of current resilience interventions.

Current resilience interventions

A substantial body of work has emerged since approximately 2010, aimed at developing and evaluating resilience interventions across various populations. The effects of these interventions have been examined in numerous reviews, 13 of which we identified as high-quality systematic reviews (published between 2014 and 2025) that evaluated interventions designed to promote resilience in the general population. We classified a review as high quality if it conducted a systematic literature search across three or more databases and adhered to standards outlined by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses³⁵ (PRISMA), the Cochrane Collaboration³⁶ and/or the Joanna Briggs Institute³⁷. We detail the main findings of the reviews and identify their limitations.

Effects across age groups

Twelve out of 13 reviews assessed post-intervention outcome differences between intervention and control groups. Three reviews investigated digital resilience interventions specifically^{27,28,38}, but the majority of interventions were delivered face-to-face, either individually or in group formats. Five reviews explored effects in adult studies (Table 1), whereas five explored studies in children and adolescents (Table 2), and three reviews examined mixed populations including children and adolescents as well as adults (Table 3). The number of primary studies included in each review ranged between 9 and 268.

Among adult populations^{26,27,29,38,39}, interventions showed consistent small (standardized mean difference, SMD = 0.20–0.49) to medium (SMD = 0.50–0.79) favourable effects of both face-to-face and digital interventions. For example, resilience interventions improved self-reported resilience (SMD = 0.37³⁹ to 0.48²⁹), symptoms of mental distress (SMD = 0.20³⁸ to 0.51³⁹), and positive mental health (SMD = 0.18³⁸ to 0.34³⁹). However, these effects were often smaller at short-term follow-up assessments 3–6 months post-intervention, and long-term effects beyond six months were rarely investigated.

Several reviews explored effect modifiers but results were inconsistent, with the exception of control group type: the favourable effect of interventions was higher when compared to passive control groups (for example, no intervention or waitlist controls) than when compared to active controls. For example, in a meta-analysis on digital resilience interventions²⁷, $SMD = 0.30$ for mental distress and 0.38 for positive mental health for the no intervention or waitlist controls, versus $SMD = 0.08$ for mental distress and 0.07 for positive mental health for the high-intensity active controls. This comparison was significant in three reviews^{27,28,38} ($P \leq 0.019$) and observed on a trend level in one review³⁹ in which only four original studies employed active control groups.

Compared to studies in adult populations, effect sizes were generally smaller and results were more heterogeneous across children and adolescent studies^{40–44}. No systematic review and only a small number of primary studies (for example, 4 of 13 studies⁴⁰) focused on children aged 12 years and younger, and findings on mental distress, positive mental health outcomes and resilience factors were mixed^{42,43} and included very small unfavourable effects on anxiety symptoms ($SMD = 0.13$) in one review⁴¹ for children and adolescents. These patterns of smaller and heterogeneous effects in children and adolescents are consistent with evidence from low-intensity psychosocial and psychotherapeutic interventions that show smaller effects at younger ages^{45–48}.

Evidence for larger beneficial effects ($SMD = 0.35–0.86$) in young age groups stemmed from a network meta-analysis⁴⁹, a type of meta-analysis that compares three or more interventions simultaneously by combining direct (head-to-head) and indirect evidence within a connected network. This analysis found that diverse approaches – such as physical activity interventions, psychotherapy, mindfulness-based interventions, skills training and psychoeducation – were effective in enhancing self-reported resilience compared to control groups. However, the studies examined included samples of young adults (age 20–25 years), psychotherapy and physical activity programmes – approaches not typically labelled as ‘resilience interventions’ – and did not synthesize effects on outcomes beyond self-reported or proxy-reported (reported by parents or teachers) resilience.

Conceptual and methodological limitations

Despite the large number of primary studies, problems in the conceptualization of resilience interventions, outcome and stressor operationalization and assessment, and trial design limit the interpretation of the reviewed research^{30,50,51}.

The systematic reviews did not use a consistent definition to select primary studies, mainly because a universally accepted definition of what constitutes a ‘resilience intervention’ is yet to be established^{27,30}. Some systematic reviews adopted the labels provided by the primary study authors²⁶, whereas others included interventions that assessed resilience (usually self-reported) as an outcome^{28,49,52,53}. A broader approach to ‘resilience interventions’ also considered interventions based on related concepts, such as stress inoculation or management, posttraumatic growth and hardiness^{27,39,54,55}.

Both lifetime stressor histories – particularly overwhelming or early-life stressors⁹ – and current perceived exposure to stressors can influence intervention effects^{56,57}. Preliminary evidence suggests that more severe exposure to adverse childhood experiences is associated with poorer outcomes following psychosocial interventions^{57,58}. However, studies examining whether exposure to various types of stressor moderates the effects of resilience interventions remain largely absent, given that exposures were neither systematically measured nor examined as potential moderators. Consequently,

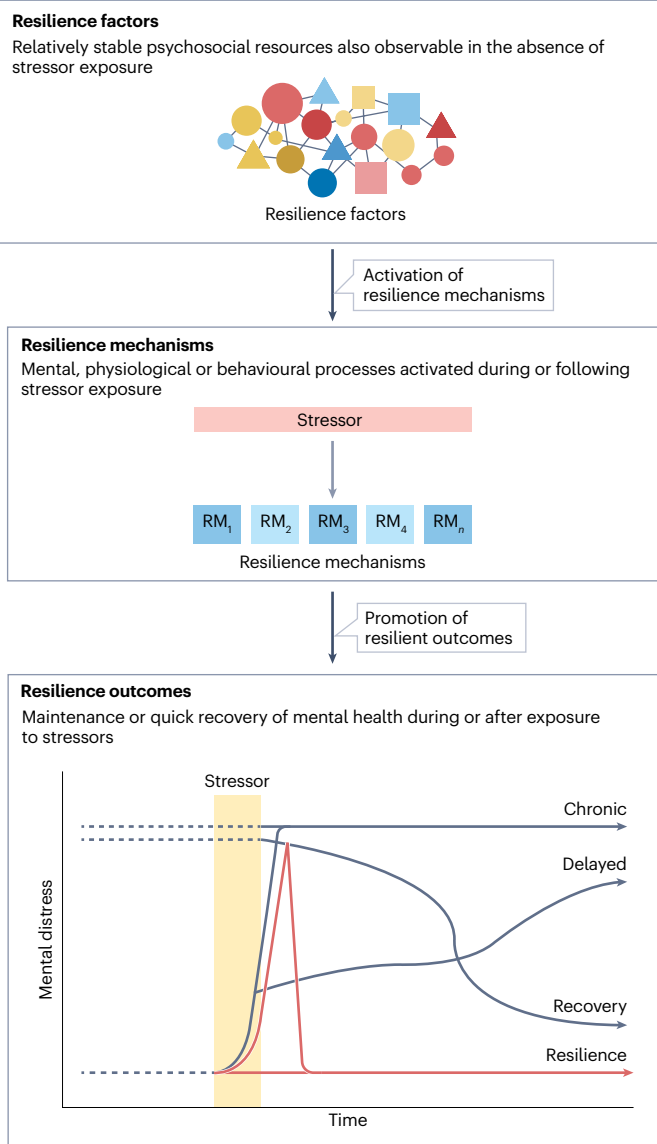


Fig. 1 | Theoretical framework of resilience processes. A broad set of resilience factors (top) increase an individual’s capability to activate a relatively smaller array of mediating resilience mechanisms (middle), which in turn facilitate resilient outcomes (bottom). Resilience factors are characteristics or resources that are stable (or change only gradually) and can therefore be assessed in the absence of or during stressor exposure¹⁹, whereas resilience mechanisms are mental, physiological or behavioural processes that come into operation only when individuals are exposed to stressors (and can therefore be measured only during stressor exposure). Resilient outcomes (red) are defined as either sustained low levels of psychopathological symptoms or rapid recovery after a temporary elevation. High and stable levels of psychosocial functioning or wellbeing – or rapid recovery of these positive mental health indicators – also represent resilient outcomes, although these have been investigated less frequently in the literature. Adapted from ref. 18, Springer Nature Limited.

insights into which interventions are most effective for individuals with differing types and degrees of past and ongoing stressor exposure remain limited.

Table 1 | Reviewed systematic reviews of resilience interventions for adults

Type of resilience intervention	Number and type of included primary studies (participants)	Group comparisons	Main post-intervention outcome ^a	Effect estimates SMD (95% CI)	Effect sizes according to Cohen ^b	Ref.
Any intervention to foster resilience or related concepts, such as hardiness or posttraumatic growth	221 RCTs (1,589–5,745 depending on the outcome), 147 passive control groups	Intervention versus control groups, differential intervention effects in Eastern and Western countries	Self-reported resilience ^c	0.48 (0.28, 0.67)	Medium	29
			Anxiety	–0.37 (–0.57, –0.18)	Small	
			Depressive symptoms	–0.46 (–0.62, –0.29)	Small	
			Quality of life	0.29 (0.15, 0.43)	Small	
			Perceived stress	–0.42 (–0.62, –0.22)	Small	
			Social support	0.05 (–0.44, 0.54)	No evidence	
Any programme designed to develop, enhance or improve resilience	11 RCTs (24–320 included in meta-analysis), 9 passive control groups	Intervention versus control groups	Self-reported resilience	0.44 (0.23, 0.64)	Small	26
Any programme designed to develop or enhance resilience or related constructs, such as hardiness	13 RCTs (782 included in meta-analysis), 16 passive control groups	Trauma-focused intervention versus control groups (3 RCTs)	Stress	–0.53 (–1.04, –0.03)	Medium	39
			Depressive symptoms	–0.51 (–0.92, –0.10)	Medium	
			Anxiety symptoms	–0.61 (–1.54, 0.31)	No evidence	
		Generalized resilience interventions versus control groups (13 RCTs)	Self-reported resilience	0.37 (0.18, 0.57)	Small	
			Quality of life	0.34 (–0.03, 0.72)	No evidence	
			Self-efficacy	0.26 (–0.10, 0.63)	No evidence	
			Stress	–0.28, (–0.60, 0.04)	No evidence	
			Depressive symptoms	–0.28 (–0.56, 0.01)	No evidence	
			Anxiety symptoms	–0.11 (–0.41, 0.20)	No evidence	
Interventions aimed at fostering reappraisal as a key resilience factor, delivered via digital devices	30 RCTs (3,904), 23 passive control groups	Intervention versus control groups	Aggregated effects (overall mental health)	0.34 (0.12, 0.56)	Small	38
			Anxiety symptoms	0.33 (0.16, 0.50)	Small	
			Depressive symptoms	0.51 (0.30, 0.73)	Medium	
			General psychological distress	0.20 (–0.25, 0.66)	No evidence	
			Wellbeing	0.18 (–0.05, 0.42)	No evidence	
Psychosocial digital interventions that either aimed at promoting resilience, hardiness, or stressor-related growth, mentioned those concepts as relevant theoretical background or addressed those concepts as outcomes	101 RCTs (20,010), 58 passive control groups	Intervention versus control groups	Mental distress	–0.24 (–0.31, –0.18)	Small	27
			Positive mental health	0.27 (0.13, 0.40)	Small	
			Resilience factors (such as self-efficacy or optimism)	0.31 (0.21, 0.41)	Small	

^aPost-intervention assessments took place within days or weeks after the end of the intervention. ^bEffect size interpretation according to Cohen²¹⁴ in this table: SMD < 0.20, very small significant average favourable effect; 0.20 ≤ SMD < 0.50, small significant average favourable effect; 0.50 ≤ SMD < 0.80, medium significant average favourable effect; SMD ≥ 0.80, large significant average favourable effect. ^cSelf-reported resilience was typically assessed using measures such as the Brief Resilience Scale⁶⁰ or the Connor–Davidson Resilience Scale⁶¹ across studies. CI, confidence interval, RCT, randomized controlled trial; SMD, standardized mean difference.

Intervention studies often relied on self-report measures based on trait-oriented conceptualizations of resilience as primary outcome^{49,59}. Self-report measures such as the Brief Resilience Scale⁶⁰ or the Connor–Davidson resilience scale⁶¹ adopt a trait-like definition of resilience. The outcomes most frequently examined are mental distress and positive mental health^{27,59}, typically assessed at post-intervention or follow-up. However, these indicators do not constitute validated measures of resilient outcomes^{25,30,50}. Conceptualizations of resilience as the maintenance or rapid recovery of mental health during or after stressor exposure require longitudinal assessment of both mental health outcomes and stressor exposure⁶², as well as analytic approaches that relate mental health status to levels of stressor exposure over time. Although methodological approaches to capture these dynamic processes have been proposed^{25,63}, they have rarely been implemented in intervention research and, consequently, are largely absent from existing systematic reviews.

In terms of trial design, a large proportion of the original studies were statistically underpowered to establish intervention efficacy or effectiveness^{26,30} and to examine potential moderators. Additionally,

evaluation studies for single resilience interventions (such as Resilience Gym⁶⁴ or SUPPORT Coach⁶⁵) remained scarce, with many interventions being tested in single, non-replicated trials^{27,59}. Consequently, there is at present no evidence that any single resilience intervention is superior to others. Furthermore, there is a paucity of studies using active control conditions, and a debate persists regarding what constitutes an ecologically valid and rigorous active comparator for resilience interventions.

Limitations in mechanistic understanding

Current theories of resilience favour process-oriented and outcome-oriented definitions of resilience over trait-oriented approaches. However, to our knowledge, only six primary studies^{66–71} and a single qualitative systematic review⁷² investigated the processes by which resilience interventions or psychosocial interventions designed for stressor-exposed populations exert their effects. To explore these processes, primary studies relied on mediation models according to which resilience interventions affect resilience factors (such as trait-level differences in positive appraisal style⁷⁰, psychological flexibility⁶⁶, positive emotions⁶⁷, mindfulness⁷¹, self-efficacy,

Table 2 | Reviewed systematic reviews of resilience interventions for children and adolescents

Type of resilience intervention	Number and type of included primary studies (participants)	Group comparisons	Main post-intervention outcome ^a	Effect estimates SMD (95% CI)	Effect sizes according to Cohen ^b	Ref.
Face-to-face group intervention to teach cognitive–behavioural and social problem-solving skills (Penn Resiliency Programme)	9 RCTs (4,744 aged 7–18 years), 6 passive control groups	Intervention versus control groups	Depressive symptoms	–0.23 (–1.09, 0.62)	No evidence	41
			Anxiety symptoms	0.13 (0.00, 0.26)	Very small (unfavourable)	
School-based resilience interventions (no specific mode of delivery) for students	38 RCTs (15,730 aged 6–19 years), 17 passive control groups	Intervention versus control groups	Self-reported resilience ^c	0.17 (0.06, 0.29)	Very small	42
School-based interventions targeting multiple protective factors supposed to ‘build’ resilience	57 RCTs (41,521 aged 7–18 years), type of control group not reported	Intervention versus control groups	Depressive symptoms	–0.08 (–0.14, –0.01)	Very small	43
			Anxiety symptoms	–0.14 (–0.28, 0.00)	No evidence	
			Hyperactivity	–0.07 (–0.18, 0.05)	No evidence	
			Conduct problems	0.01 (–0.11, 0.12)	No evidence	
			Internalizing problems	–0.21 (–0.36, –0.06)	Small	
			Externalizing symptoms	–0.18 (–0.34, –0.01)	Very small	
			General mental distress	–0.11 (–0.21, –0.01)	Very small	
School-based interventions targeting at least one environmental and one individual protective factor to reduce adolescent tobacco, alcohol or illicit substance use	19 RCTs (51,867), 29 passive control groups	Intervention versus control groups	Tobacco use	OR=0.96 (0.85, 1.08)	No evidence	44
			Alcohol use	OR=0.86 (0.73, 1.02)	No evidence	
			Illicit substance use	OR=0.78 (0.66, 0.93)	Small	
Any intervention to foster resilient outcomes using psychological and/or psychotherapeutic methods	13 RCTs (27–2,996 ≤22 years, 4 RCTs reporting on children), 10 passive control groups	Intervention versus control groups	Self-reported resilience	0.48 (0.15, 0.81)	Small	40
			For short-term (≤3 months) and mid-term follow-ups (3–6 months): no evidence for favourable effects			

^aPost-intervention assessments took place within days or weeks after the end of the intervention. ^bEffect size interpretation according to Cohen²¹⁴ in this table: SMD < 0.20, very small significant average favourable effect; 0.20 ≤ SMD < 0.50, small significant average favourable effect; 0.50 ≤ SMD < 0.80, medium significant average favourable effect; SMD ≥ 0.80, large significant average favourable effect. ^cSelf-reported resilience was typically assessed using measures such as the Brief Resilience Scale⁶⁰ or the Connor–Davidson Resilience Scale⁶¹ across studies. CI, confidence interval; OR, odds ratio; RCT, randomized controlled trial; SMD, standardized mean difference.

Table 3 | Reviewed systematic reviews of resilience interventions for mixed populations from children to adults

Type of resilience intervention	Number and type of included primary studies (participants)	Group comparisons	Main post-intervention outcome ^a	Effect estimates SMD (95% CI)	Effect sizes according to Cohen ^b	Ref.
Any training of personal and relational protective factors (such as cognitive flexibility or self-efficacy) delivered via digital devices	22 RCTs (2,876), 19 passive control groups	Intervention versus control groups	Self-reported resilience ^c At follow-up assessment: Large favourable average effect for resilience	0.54 (0.28, 0.79)	Medium	28
Any intervention aiming at the promotion of resilience	268 trials (26,315 of all ages), control groups not considered	Pre-to-post intervention changes	Aggregated effect	0.48 (0.40, 0.56)	Small	59
			Action	0.76 (0.34, 1.18)	Medium	
			Biophysical	0.08 (-0.04, 0.20)	No evidence	
			Coping	0.43 (0.30, 0.57)	Small	
			Emotion	0.57 (0.21, 0.93)	Medium	
			Self-reported resilience	0.44 (0.35, 0.54)	Small	
			Mental distress	0.51 (0.40, 0.61)	Medium	
			Wellbeing	0.41 (0.33, 0.49)	Small	
Any intervention labelled as 'resilience-promoting' for adolescents and young adults (including physical activity interventions and psychotherapy)	46 RCTs (8,729 adolescents and young adults), 32 passive control groups	Network meta-analysis comparing different intervention types versus treatment as usual	Self-reported resilience	Physical activity: 0.86 (0.30, 1.42)	Large	49
				Psychotherapy: 0.76 (0.45, 1.07)	Medium	
				Mindfulness: 0.48 (0.19, 0.78)	Small	
				Skills training: 0.43 (0.21, 0.65)	Small	
				Psychoeducation: 0.35 (-0.16, 0.86)	Small	

^aPost-intervention assessments took place within days or weeks after the end of the intervention. ^bEffect size interpretation according to Cohen^{71a} in this table: SMD < 0.20, very small significant average favourable effect; 0.20 ≤ SMD < 0.50, small significant average favourable effect; 0.50 ≤ SMD < 0.80, medium significant average favourable effect; SMD ≥ 0.80, large significant average favourable effect. ^cSelf-reported resilience was typically assessed using measures such as the Brief Resilience Scale⁶⁰ or the Connor-Davidson Resilience Scale⁶¹ across studies. CI, confidence interval; RCT, randomized controlled trial; SMD, standardized mean difference.

positive thinking, altruism and goal setting⁶⁹), and changes in these factors drive intervention effects (improved outcomes). Despite encouraging findings in single studies, none of the studied processes have yet been replicated, leaving the question of what causes favourable effects of resilience interventions unanswered.

In examining these processes, studies typically relied on linear mediation models within randomized controlled trial designs. In such models, the intervention (independent variable) is hypothesized to influence the outcome (dependent variable) indirectly through a mediator. Full mediation is inferred if the direct path from the intervention to the outcome becomes non-significant⁷³. These models often include a temporal component, assuming that the intervention precedes the mediator, and the mediator precedes the outcome. However, this modelling approach has been increasingly criticized^{73–75}. First, mediation models make assumptions at the group level, but the paths observed at the group level might not translate to or might even be inverted at the individual level⁷³. Second, they assume unidirectional and temporally stable relationships among variables, which might not reflect the dynamic nature of psychological change⁷⁶. Third, these models cannot account for nonlinear interactions, feedback loops or the interdependence of variables over time⁷³.

The limited understanding of how resilience interventions bring about change might limit room for improvement and contribute to

the modest effect sizes observed across resilience interventions, thereby constraining progress along the later stages of the translational research pipeline, including implementation and dissemination. We delve into the rationales of these interventions in the next section.

Critical examination of rationales

Without clarity about mechanisms of change, it remains difficult to determine which intervention components are essential, for whom they work, and under what conditions they are effective. This gap becomes particularly consequential when moving from controlled research settings to real-world delivery.

Resilience factors in intervention research

Most interventions are premised on the assumption that resilience can be enhanced through the training of resilience factors, typically individual-level psychosocial resources⁵⁵. This factor-based approach reflects a historical focus of resilience research on identifying and cataloguing individual resilience factors⁷⁷. For example, a systematic review of resilience in the context of large-scale societal stressors (such as pandemics, natural disasters or armed conflicts) identified 54 distinct resilience factors across 50 studies, most of which focused on intrapersonal psychosocial characteristics (such as optimism or self-efficacy²⁰). A large-scale open access database on trajectory-based

resilience research (resilience META⁷⁸) identified 173 resilience factors across 157 primary studies, of which 102 were individual-level factors, 58 were social factors and 11 pertained to the societal level – although social factors beyond perceived social support^{79,80} (such as family functioning and social network size) and societal factors (such as trust in the legal system or perceived societal solidarity) remain underexplored^{20,81}.

There is no clear justification for selecting specific resilience factors over others in intervention research. Conceptually, many of these factors exhibit considerable overlap, making it difficult to distinguish them theoretically⁸². For example, self-efficacy and control beliefs are conceptually difficult to differentiate because both constructs centre on individual perceptions of agency over outcomes⁸³. Self-efficacy refers to beliefs about one's capability to execute actions⁸⁴, whereas control beliefs encompass expectations about whether outcomes are contingent on one's behaviour or external factors⁸⁵. However, these domains overlap substantially: judgements about one's capability are often intertwined with expectations about control over outcomes, and both are shaped by similar experiences, such as past success or failure, resulting in large correlations (for example, $r = 0.60$ ⁸⁶ or $r = 0.65$ ⁸⁷). There is no consensus on a definitive list of resilience factors, nor a unified theoretical framework that organizes them coherently.

Furthermore, interventions often target an undefined or overly broad set of factors, and testing several factors simultaneously can inflate outcome measures^{30,59}. For instance, a three-session resilience intervention for university students focused on fostering optimism and perceived personal control⁸⁸, and a similar programme for the general public trained a range of resilience factors including self-efficacy, optimism, positive emotions, coping skills, emotion regulation, social support, self-esteem, cognitive flexibility and growth mindset⁸⁹. Empirically, the predictive validity of individual resilience factors is often weak and/or inconsistent. Relevant parts of the literature rely on cross-sectional designs that document correlations between presumed resilience factors and psychological outcomes – for instance, associations between social support^{79,90}, sense of coherence^{91,92}, or self-efficacy⁹³ and lower psychological distress or better positive mental health. However, these findings do not imply causality and are vulnerable to interpretive confounds, such as conceptual redundancy⁹⁴, reverse causality (for example, psychological distress influencing resilience factors⁹⁵) or shared variance due to unmeasured third variables like stressor exposure.

Longitudinal evidence would be needed to demonstrate that specific factors consistently predict resilient outcomes. A review²⁰ solely based on longitudinal studies found that perceived social support, better emotion regulation and higher psychological flexibility were robustly associated with resilient outcomes. Still, effect sizes were categorized as 'small', and the variance in resilient outcomes remaining unexplained was substantial – even after aggregating the predictive value of multiple resilience factors^{87,96–98}. Robust longitudinal designs that incorporated pre-stressor mental health baselines also yielded inconsistent findings. These studies showed that the predictive value of resilience factors varied across stressor types and within the same stressor across contexts^{18,20,77,99}, suggesting that the relevance of specific resilience factors is context-dependent^{18,77}. Other studies revealed paradoxical findings; for example, higher levels of certain presumed resilience factors (control beliefs, supportive coping, positive reframing) were associated with non-resilient outcomes for the same stressor^{87,100,101}.

Conceptual foundations and target populations

Many resilience interventions lack a clear theoretical underpinning or a well defined theory of change, resulting in substantial gaps in conceptual foundation^{30,40,51}. For training resilience factors, interventions frequently draw from established psychotherapy approaches, such as cognitive-behavioural therapy¹⁰², mindfulness-based interventions¹⁰³, acceptance and commitment therapy¹⁰⁴, or stress inoculation training¹⁰⁵. These interventions are typically delivered through a variety of exercises also known from psychotherapy, including psychoeducation, relaxation techniques, cognitive restructuring and role-play scenarios^{54,55}. The theoretical rationale for these exercises depends on the intervention's underlying framework, which generally emphasizes learning processes at the behavioural, social and cognitive level. From this perspective, resilience can be promoted either by addressing learning patterns that reduce the chance of resilient outcomes (for example, modifying dysfunctional appraisals) or by strengthening learning processes that enhance the chance of resilient outcomes (for example, fostering active coping or the effective use of social resources). These pathways are assumed to interact dynamically; for example, reductions in maladaptive appraisals might facilitate the uptake and effectiveness of adaptive coping strategies, whereas the successful application of such strategies can, in turn, reinforce functional appraisals over time, potentially creating a reinforcing cycle that promotes resilient outcomes. Those learning processes are paralleled by intervention-related changes at the neurobiological level¹⁰⁶. Together with contextual and environmental factors (Box 1), these processes produce the often small-to-medium favourable effects of resilience interventions.

Interventions should be selected according to the individuals that need them, but target populations are generally poorly defined in intervention studies. Target populations are groups assumed to be at elevated risk for stressor exposure (for example, adolescents and young adults¹⁰⁷, individuals with very demanding workplaces^{108,109} or those facing major life events such as recent diagnoses of serious illness¹¹⁰) or identified by self-selection strategies (for example, individuals from a general working population willing to participate in a study on resilience¹¹¹). Populations less likely to engage with mental health services, such as middle-aged to older men¹¹² and individuals from socioeconomically disadvantaged or marginalized backgrounds¹¹³, remain underrepresented in intervention research. Thus, resilience intervention studies often fail to adequately reach those who stand to benefit most – individuals exposed to high levels of stress within low-resource contexts^{114–116}.

Implementation of interventions

So far, research into resilience interventions broadly employs a linear translational research pipeline^{117,118} starting with researcher-led intervention development, followed by efficacy and effectiveness trials before implementation and dissemination research starts, and ending with evaluation in real-world settings¹¹⁹ (Table 4). However, given the small effect sizes found for many of these interventions, it remains unclear whether these effects might be amplified or rendered negligible in real-world contexts, which is more likely thanks to the so-called 'voltage drop'¹²⁰ (the tendency for intervention effects observed under controlled research conditions to diminish when translated into routine practice). These limitations create a critical knowledge gap concerning effective strategies for scaling and deploying interventions^{114,121}.

Furthermore, efficacy and effectiveness studies are typically conducted with privileged populations^{122,123} in high-resource contexts^{27,114,124}. Interventions are usually developed, tested and

Box 1 | Multisystem and contextual interventions to promote resilience

Resilience emerges from dynamic interactions across multiple systems, including individuals, families, communities, institutions and broader sociocultural and structural environments³³. Multisystem frameworks in resilience science highlight that contextual and environmental systems have roles in shaping responses to adversity at least as important as those of individual-level systems²¹⁵ — and in contexts such as natural disasters, forced displacement or community-level violence their roles might be even more relevant. Incorporating these broader systems into resilience interventions research offers critical opportunities for advancing mechanism-focused approaches.

Contextual systems

Community and societal environments shape the availability, accessibility and mobilization of resources that enable resilient outcomes. Key contextual systems are:

- Social and community resources: social networks, community organizations, collective efficacy, shared norms, peer support and informal helping structures²¹⁶
- Institutional and service systems: schools, healthcare systems, mental-health and social services, emergency response infrastructure and local governance³³
- Structural and policy environments: housing stability, economic security, transportation, neighbourhood safety, and public policies influencing education, health, labour and social protection²¹⁷

Interventions targeting these systems aim to strengthen the external conditions that individuals rely on when exposed to stressors. Such efforts can reduce exposure to ongoing disadvantages, improve access to resources, and promote environments conducive to recovery and growth²¹⁶.

Environmental systems

Multisystem and contextual responses are often required to protect populations and reduce secondary harms¹⁷⁸. Interventions at this level target resilience mechanisms by modifying environmental conditions²¹⁸. Although usually not labelled as ‘resilience interventions’, some examples are^{216,219}:

- Timely evacuation and relocation after natural disasters, thus reducing exposure to danger and facilitating rapid mobilization of support
- Rebuilding schools, community centres and local infrastructure, which restores continuity, safety and social connection
- Coordinated public health and social-service responses, such as provision of food, shelter, financial assistance and healthcare
- Climate adaptation strategies, such as community preparedness planning, heat-mitigation policies or disaster-resistant housing

Complementarity with individual-focused approaches

Contextual and environmental interventions do not replace individual-level resilience interventions but have the potential to substantially strengthen them. For example, individual resources such as emotion regulation strategies might be insufficient in environments marked by chronic instability, low resources or structural deprivation²¹⁵.

Integrated interventions that coordinate supports across levels might therefore yield more sustainable benefits, especially for populations facing chronic adversity or intersecting vulnerabilities. Strengthening both individuals and the systems surrounding them might substantially enhance the public-health impact of resilience interventions^{215,220}.

disseminated among lower-risk, higher-resource groups, whereas individuals facing substantial stressor exposure remain largely unreached. Resilience interventions might show promising effects in privileged samples but contribute little to reducing the broader burden of stressor-related problems if they lack scalability and might risk exacerbating existing structural health inequalities^{114,125}.

This distributional mismatch echoes the well known ‘prevention paradox,’ which states that preventive interventions might yield the greatest population-level benefit when delivered broadly, although the individuals at highest risk would, in principle, benefit most from targeted support^{126,127}. At the same time, those at highest risk are often the hardest to reach with preventive measures, because they face more substantial structural, social and psychological barriers to accessing such programmes¹²⁸ — a phenomenon known as the ‘prevention dilemma’¹²⁹.

Implementation research has introduced various frameworks to systematically evaluate these processes, among which a heuristic taxonomy^{130,131} is one of the most widely adopted¹³². This framework identifies eight key implementation outcomes: acceptability, feasibility, fidelity, implementation costs, sustainability, appropriateness, penetration and adoption (Box 2). So far, no systematic review has yet focused on the implementation outcomes of resilience interventions, mainly because primary studies assess only a subset of early implementation outcomes — usually acceptability, feasibility and fidelity — and

often only as secondary focus^{133–135}, with primary attention given to intervention effects. Whereas some studies have addressed implementation costs and cost-effectiveness for individual interventions¹³⁶, there is almost no evidence on appropriateness, penetration and adoption — outcomes that are particularly critical for understanding how interventions perform when scaled to broader populations.

The lack of sustainability might underlie this knowledge gap. The predominance of single-trial evaluations probably reflects the broader issue that most interventions do not persist beyond the funding period of individual research projects. For instance, a review of digital resilience interventions²⁷ assessed the ongoing availability of web- and app-based programmes evaluated in primary studies. Searches of websites and app stores revealed that only a small number of these interventions remained publicly accessible.

In summary, the empirical foundation for many individual-level resilience interventions remains limited, reflecting a prevailing emphasis on resilience factors that show only weak or inconsistent associations with resilient outcomes^{18,77}. At the same time, relatively little research has focused on the processes that generate intervention effects. This gap — longstanding in psychotherapy research⁷³ — reflects a broader tendency to prioritize outcomes over the mechanisms that produce them. Identifying these change processes might not only clarify how resilience interventions work but also provide ways to enhance their

magnitude and sustainability¹³⁷. Given this scenario, in the next sections we re-evaluate the current prevailing conceptual framework through the lens of a growing body of research focused on resilience mechanisms.

From factors to mechanisms

The broader field of resilience research is increasingly focusing on identifying the processes connecting resilience factors with resilient outcomes. Integrating these broader theoretical frameworks and empirical approaches in resilience interventions research could promote the development of interventions that primarily target mechanisms and overcome the field’s current limitations.

Theoretical frameworks

Two current prominent theories of resilience mechanisms are regulatory flexibility^{18,138} and positive appraisal style^{19,25}. The concept of regulatory flexibility^{18,138} is grounded in the premise that coping strategies are not inherently adaptive or maladaptive; rather, their effectiveness depends on contextual fit. For example, seeking emotional support might be ineffective if it is unavailable at a particular time or in a specific context, potentially delaying the adoption of other more fitting strategies such as acceptance or cognitive reappraisal. Thus, regulatory flexibility emphasizes the dynamic process of matching strategies to varying situational demands¹³⁸. This theoretical framework extends prior work on emotion regulation flexibility¹³⁹. Whereas emotion regulation flexibility focuses primarily on the covariation between regulatory strategies and situational features – that is, the outcome of successful fit – regulatory flexibility additionally encompasses the processes involved in strategy selection, implementation and adjustment, alongside the resulting fit¹⁴⁰. Regulatory flexibility involves stable resilience factors – such as having a flexibility mindset (providing the motivation to flexibly change behaviour, and comprises beliefs such as optimism and coping self-efficacy^{18,77}), possessing a rich repertoire

of potential coping strategies, being effective in their selection and employing effective monitoring strategies¹³⁸ – that provide the basis for the resilience mechanism that comes into play when coping with acute stressors. These processes ensure the implementation of contextually sensitive strategies and switching those when needed.

The second theory is the positive appraisal style theory (PASTOR²⁵), which posits that the capacity to optimally respond to stressors is compromised under conditions of overwhelming stressor-induced aversive motivation. Within this framework, the stable resilience factors are individual appraisal tendencies characterized by a consistent bias towards mildly (but not unrealistically) positive appraisals, that is, a slight underestimation of negative outcome costs (trivialization) and negative outcome probabilities (optimism) and a slight overestimation of one’s coping potential (control, confidence), collectively termed a ‘positive appraisal style’¹⁴¹. Such a positive appraisal style makes it more likely that an individual will generate positive appraisals during stressor exposure (the resilience mechanism), which in turn facilitates resilient outcomes.

Despite having been prominently discussed in resilience research^{19,20,23}, empirical support for these two theories remains preliminary. A review of resilience factors in the context of large-scale societal stressors²⁰ found consistent support for flexibility-related constructs. However, the literature encompasses only partly overlapping conceptualizations of ‘flexibility,’ including coping flexibility (the ability to vary and adapt coping strategies in response to situational demands¹⁴²), and psychological flexibility (the ability to respond to situations in ways that facilitate valued goal pursuit¹⁴³), which reflect flexibility-related resilience factors rather than the resilience mechanism. Cross-cultural evidence for all components of regulatory flexibility (context sensitivity, repertoire and feedback responsiveness) comes from a study¹⁴⁴ conducted in China and the USA that tracked individuals over three weeks using ecological momentary assessment.

Table 4 | Translational research pipeline

Research pipeline	Name	Focus	Type of research	Study design	Key actors	Current priority in resilience research
Stage 1	Broad resilience research	Identifying intervention targets and mechanisms	Research of resilience factors and outcomes	Experimental and observational studies	Researchers	Very high
Stage 2	Efficacy research	Studying intervention effects under highly controlled, nearly ideal conditions	Intervention research	Randomized controlled trials	Researchers	High
Stage 3	Effectiveness research	Assessing intervention effects in more realistic settings, still equipped for high-fidelity implementation	Intervention research	Randomized controlled trials, including pragmatic trials	Researchers, less often implementators	Medium to high
Stage 4	Implementation and dissemination research	Distributing evidence to target groups (such as policy-makers and community leaders), implementation in real-world settings	Implementation research	Mixed methods designs, and focus on qualitative data	Researchers, implementators, stakeholders and participants	Low
Stage 5	Routine implementation or scaling up	Delivering interventions in intended settings and populations under routine conditions	Practice or service implementation	Observational studies in routine service delivery, scale-up initiatives and mixed methods	Implementators, stakeholders and participants	Almost absent
Stage 6	Real-world evaluation or monitoring	Assessing reach, uptake, fidelity, outcomes and sustainability in routine practice	Evaluation research	Naturalistic observational studies, mixed methods and monitoring data	Researchers, implementators, stakeholders and participants	Almost absent

Box 2 | Outcomes of implementation studies

The heuristic taxonomy of implementation outcomes as proposed by Proctor and colleagues^{130,131} describes eight implementation outcomes. We provide a brief description of these outcomes on the basis of a systematic review on implementation outcomes of psychosocial interventions for children and adolescents living in low- and middle-income countries, including resilience interventions²²¹.

Acceptability

The extent to which stakeholders perceive the intervention as satisfactory and agreeable. This outcome is evaluated at the level of providers and participants.

Feasibility

The practicality of successfully delivering the intervention. This outcome is assessed at the level of providers, organizations or settings.

Fidelity

The degree to which the intervention is delivered as originally designed. This outcome is assessed at the level of providers.

Implementation costs

The total expenses associated with delivering the intervention. This outcome is assessed at the level of providers or implementing institutions.

Sustainability

The continued use and maintenance of the intervention over time. This outcome is assessed at the level of administrators, organizations or settings.

Appropriateness

The perceived relevance, fit and compatibility of the intervention with the specific context, providers and users. This outcome is assessed across all actors and levels: providers, participants, organizations and settings.

Penetration

The extent to which the intervention is integrated into real-world practice and reaches its intended target population. This outcome is assessed at the level of the organizations or settings.

Adoption

The initial uptake and implementation of the intervention in real-world settings. This outcome is evaluated at the level of providers, organizations or settings.

Higher context sensitivity and a broader repertoire were consistently and independently associated with lower momentary distress, whereas findings for feedback responsiveness were more heterogeneous.

Evidence from two additional longitudinal studies during the first year of the COVID-19 pandemic^{145,146} further indicates that individuals with a broader repertoire of coping strategies showed more favourable trajectories of anxiety and depression. A prior study found that a broader repertoire prospectively predicted better long-term mental health following terror attacks¹⁴⁷. Complementarily, in three cross-sectional studies with non-clinical community samples, individuals with a broader repertoire tended to experience lower levels of psychological distress^{148–150}, and changes in psychological flexibility were a potential mediator of positive intervention outcomes in stressor-exposed populations in another study⁶⁶ (but see ref. 67). This finding suggests that flexibility-related concepts might constitute a mechanism through which resilience interventions exert their effects.

Similar evidence also supports the positive appraisal theory. Higher levels of positive appraisal style were repeatedly associated with resilient outcomes in cross-sectional^{151,152} and longitudinal studies^{70,153–155}. Higher levels of positive appraisal style have also been shown to mediate the effects of resilience factors on resilient outcomes during the COVID-19 pandemic^{70,151,152,154,155} and of a low-intensity transdiagnostic psychosocial intervention in stressor-exposed populations⁷⁰. This latter finding suggests that appraisal processes might play a key part in the mechanisms of change during resilience interventions.

In summary, although current evidence does not yet justify designating positive appraisal style or regulatory flexibility as overarching mechanisms of resilience, emerging findings provide a promising foundation for further investigation.

Applying theory to resilience interventions

Resilience interventions explicitly targeting the mechanisms outlined have been proposed^{18,156} but have yet to be empirically evaluated. Grounded in the notion that resilience mechanisms are activated during acute stressor exposure^{19,23}, interventions aimed at modifying resilience mechanisms might involve training individuals under conditions of acute stress or through simulations that mimic real-world stressor exposure. According to the regulatory flexibility framework^{18,138}, interventions might support individuals in the flexible selection and implementation of coping strategies according to situational demands as well as in adapting their behaviour in response to feedback. According to the positive appraisal style theory^{19,25}, interventions might enhance mild positive appraisal or reappraisal processes during acute stressor exposure¹⁵⁶.

To date, just-in-time adaptive interventions triggered by ecological momentary assessments and delivered during acute stressor exposure have been found to be feasible¹⁵⁶, although evidence regarding their effects remains lacking. In contrast, simulation-based training has primarily been employed in military contexts^{157–159}. Such training could be adapted to more heterogeneous civilian settings (for example, stressor-prone workplaces) to enable the modification of mental and behavioural processes during simulated stressor exposure, and could be combined with intervention content aimed at strengthening more stable psychosocial resilience factors that facilitate the recruitment of resilience mechanisms. For example, under the regulatory flexibility framework^{18,138}, exercises might promote a flexible mindset or enlarging one's repertoire of coping strategies, whereas under the positive appraisal style theory^{25,19}, interventions might enhance an individual's general, stable tendency towards positive appraisal.

Empirical insights from transdiagnostic research

Empirical insights into the potential effectiveness of mechanistic-focused interventions can be drawn from research on transdiagnostic programmes^{160,161}, targeting a broad range of mental symptoms. These programmes explicitly address individuals facing acute and often pervasive stressors (such as displacement or pandemics^{162,163}) and therefore provide a valuable model for interventions targeting diagnosis-independent processes that operate during periods of heightened stress. Central to these approaches is a focus on transdiagnostic strategies established in psychotherapy^{164,165} – such as problem or stress management¹⁶⁰, cognitive flexibility^{166,167}, emotion awareness and emotion regulation strategies^{161,166}, or the mobilization of social support¹⁶⁰. Although these strategies are typically discussed in relation to the development and persistence of diverse mental health conditions, they might be equally relevant for maintaining mental health and promoting rapid recovery during or following stressor exposure^{20,168}. When implemented under conditions of acute stressor exposure, these strategies can be conceptualized as interventions focusing on resilience mechanisms. However, despite this conceptual overlap, transdiagnostic programmes have only rarely been explicitly framed as resilience interventions^{70,169–171}.

For such interventions, a substantial body of evidence indicates that transdiagnostic approaches reduce mental distress and improve quality of life in adults exposed to stressors, with effects that are often larger and more durable than those reported for interventions explicitly designed to enhance resilience by targeting resilience factors^{45,172}.

These findings support a reconceptualization of resilience interventions, shifting the emphasis from the identification and promotion of putative resilience factors towards the direct targeting of transdiagnostic resilience mechanisms that support adaptive functioning under acute stress. At the same time, pronounced age-related variability^{45,48,172} in intervention effects underscores the importance of specifying under which developmental conditions such mechanism-focused interventions are most likely to be effective.

Adopting a developmental perspective

Resilience – and the factors and mechanisms that support it – develops across the lifespan^{173,174}, with critical learning processes and changes occurring during childhood and adolescence¹⁷⁵. Thus, resilience interventions must adopt a lifespan-sensitive perspective that accounts for developmental differences in stressor exposure, coping capacities, and the neurobiological, cognitive and social processes underlying resilience^{33,176} as well as differences in the phenotypic expression of resilient outcomes across the lifespan¹⁷⁷.

Resilience interventions in childhood and adolescence might capitalize on developmental windows of opportunity¹⁷⁸. Developmentally informed approaches conceptualize human development as unfolding through dynamic interactions across multiple levels, from genetic and neurobiological processes to social and cultural influences¹⁷⁹. Within this framework, resilience emerges from the functioning of adaptive systems that operate both within individuals (for example, stress reactivity and self-regulation systems) and across interconnected contexts, including caregiving relationships, family environments, schools and broader communities¹⁷⁷. These systems also give rise to key individual-level resilience mechanisms such as positive appraisal style or regulatory flexibility.

From this perspective, resilience interventions might operate by modifying these underlying mechanisms within and across systems, rather than by targeting isolated psychosocial resilience factors with a

focus on the individual level. Windows of opportunity for intervention are thought to occur at key developmental transitions and under conditions of stressor exposure, when these systems are particularly malleable (for example, during school entry or following family disruption)^{177,180}. These periods are marked by heightened plasticity in mechanisms underlying resilient outcomes, including appraisal processes and regulatory flexibility. Aligning interventions with these windows of opportunity is therefore likely to enhance their impact, with early, multisystem interventions holding particular promise owing to their potential to initiate cascading effects across both life domains and the lifespan³³.

Consistent with this view, emerging evidence suggests that developmental stage moderates intervention effects. Findings of both resilience interventions⁴⁰ and transdiagnostic programmes targeting stressor-exposed populations¹⁶¹ were less favourable for younger populations compared to adults^{45,48}. Interventions that demonstrate favourable effects in middle-aged adult populations might not be effective in fostering resilient outcomes in younger and older individuals, for whom the traditional focus on resilience factors, often at the individual level, might be especially limited because it neglects the multisystemic nature of resilience. Intervention research that fails to account for developmental stage therefore remains inherently incomplete. Particularly in children and adolescents, such lack of attention neglects one of the most critical target groups for resilience interventions because their exposure to overwhelming stressors has been shown to have detrimental long-term effects¹⁸¹ but also because they possess substantial potential for long-term learning processes.

Modelling change

Dynamic network models are models in which a system (network) is defined by a set of nodes (variables such as resilience factors or mental health symptoms) and edges (their relationships), often represented by partial correlation coefficients. Temporal information can also be encoded, enabling researchers to identify how changes in one node prospectively predict changes in others or the same node over time. This approach is employed in psychotherapy research to study individual-level change processes over time¹⁸² and overcomes the limitations of traditional linear mediation models⁷³.

Dynamic networks are increasingly employed to model the complex interrelations among resilience factors^{87,183–185}, although only a few studies have adopted temporal network analyses^{186,187}. So far, no trials investigating resilience interventions have used this approach to model change processes. Such network models could reveal how interventions alter the structure and dynamics of relationships among resilience factors, resilience mechanisms, mental health symptoms and indicators of positive mental health⁷⁶. For example, they can show whether an intervention introduces or strengthens resilience mechanisms – thereby shifting the overall network architecture. Variables for inclusion in such networks might be drawn from prior research identifying common mediators in psychotherapy¹⁸⁸.

Dynamic network modelling requires high-frequency, time-series data, such as those obtained through ecological momentary assessment. This methodology has been used to examine associations between regulatory flexibility and affect¹⁴⁴ and is increasingly being adopted for delivering resilience interventions^{156,189}, but intervention studies have not yet focused on dynamic within-person change processes. Such studies would offer a unique opportunity to generate initial insights into mechanisms of change at the person level; for example, by repeatedly assessing intervention targets alongside ecologically valid outcomes and current levels of stressor exposure.

Moreover, pre–post comparisons of aggregated indices from ecological momentary data could be used to test whether the targeted mechanisms shift as intended and whether such shifts are associated with changes in resilient outcomes – both at the individual and group level.

Towards tailored interventions

Focusing on mechanisms in intervention research can lead to tailored resilience interventions. For example, dynamic network models capture person-level dynamic processes that can be complemented with the identification of robust, group-level change processes that generalize across individuals. Integrating these resources might enable the simultaneous identification of common resilience mechanisms and their person-level manifestations, providing a foundation for scalable yet individualized resilience interventions. In this section, we describe the variables that might influence the effectiveness of interventions across individuals.

Stressor characteristics

Stressors can be characterized according to the intensities of exposure (for example, daily hassles, repeated exposure to occupational stressors or major life events) and stressor types (for example, traumatic and non-traumatic, accidental and interpersonal)^{2–4}. These stressors differ in their duration, predictability, controllability and frequency¹⁹⁰, as well as with respect to their potential long-term negative mental health impacts^{191,192}.

Whether responses to stressor exposure are determined primarily by the type of stressor encountered, or instead by overarching, stressor-independent mechanisms such as positive appraisal style and regulatory flexibility, remains a subject of ongoing research. Some evidence supports that resilience factors and mechanisms are linked to resilient outcomes across contexts and stressor types regardless of the type or intensity of stressors. For example, emotion regulation flexibility and psychological flexibility were consistently associated with resilient outcomes across multiple stressors in a study of public health crises²⁰. Similarly, trait-level differences in positive appraisal were associated with resilient outcomes following non-COVID-related daily hassles and COVID-19-related stressors¹⁵¹. Moreover, those trait-level differences in positive appraisal were found to be linked to resilient outcomes across heterogeneous life events and daily hassles in two longitudinal observational studies in the general population⁷⁰.

By contrast, some theoretical perspectives argue that resilience is stressor-specific¹⁹⁰. Indeed, empirical studies suggest that the influence of resilience factors and mechanisms on resilient outcomes might depend on individual levels of stressor exposure¹⁹³ and exposure to moderate stressors might even have resilience-promoting effects^{179,194}.

However, the relationship between stressor exposure and the effects of resilience interventions remains poorly understood. Only a small number of intervention studies (one only published as a preprint¹⁹⁵) have incorporated outcome measures that reflect individual differences in stressor exposure, such as stressor reactivity indices^{30,70,195}. Of those that did, no study examined whether stressor exposure moderates intervention effects. Two studies reported that lifetime stressor exposure – particularly adverse childhood experiences – moderated intervention effects, such that interventions were less effective in individuals with greater lifetime exposure^{57,58}.

At present, it remains unclear to what extent resilience is stressor-dependent or consistent across heterogeneous stressor types and whether intervention effects depend on current levels of exposure – an issue with substantial implications for the design of

resilience interventions. If resilience and intervention effects prove to be stressor-dependent, a precise assessment of participants' lifetime and current stressor exposure would represent a critical starting point for targeted interventions.

Timing

The timing of resilience interventions relative to stressor exposure remains underexamined^{30,156}. Interventions delivered before exposure might be best suited to strengthening resilience factors, whereas those delivered during or shortly after exposure might more effectively target mechanisms involved in acute coping and early recovery. Optimal timing is unlikely to be uniform³⁰ and should be informed by developmental phase¹⁹⁶, stressor characteristics (for example, intensity, predictability and controllability), and contextual constraints such as the availability of social and institutional support. Developing more nuanced, developmentally informed models of intervention timing might therefore enhance the precision and effectiveness of resilience interventions.

Cultural factors

Cultural influences shape emotion-regulation strategies^{197,198} and coping behaviour¹⁹⁹. Thus, resilience interventions might operate differently in different cultural contexts. For example, in a systematic review the effects of resilience interventions were found to be more favourable in collectivistic compared to individualistic cultures²⁹. These findings underscore the importance of cross-cultural validation of resilience mechanisms and related interventions¹⁴⁴.

Despite these differences, resilience interventions are generally developed and evaluated within culturally homogeneous samples – often in Western high-income countries^{27,29} – and implicitly assume the universality of relevant targets, mechanisms and outcomes. Greater attention to cultural diversity – both between and within populations – might enhance intervention effects²⁰⁰ by improving the alignment between intervention content, targeted factors and mechanisms, and culturally embedded values, norms and social practices²⁹. Incorporating culturally informed adaptations and systematically examining cultural moderators of intervention effects represent important steps towards more effective resilience interventions.

Individual characteristics

Diversity across individuals with respect to age, gender, socioeconomic status, educational background and current levels of mental distress²⁰¹ might also influence resilience mechanisms and intervention effects. Conventional meta-analyses typically rely on aggregated data, which average between-study differences and preclude the examination of within-study or individual-level variability. However, these meta-analyses are prone to ecological bias, because, for example, between-study variations in the average age across studies might not accurately represent within-study, individual-level effects²⁰², and effects can even be inverted for between-study and within-study levels.

Consequently, the implications of findings from conventional meta-analyses for tailoring and individualizing resilience interventions are highly limited. To overcome these limitations, one-stage individual participant data meta-analyses are recommended^{36,202}. These types of meta-analysis pool individual-level data from primary trials to enable more precise, well powered analyses of individual-level moderators such as age, gender or within-study cultural diversity. However, these meta-analyses require a substantial number of sufficiently powered primary studies employing comparable interventions and providing open data that are currently not available for resilience interventions²⁷.

Summary and future directions

Moving beyond previous conceptual reviews on resilience interventions that focused mainly on intervention effects^{26,39,49,59}, our Review integrates the current research on the drivers and mechanisms of change through which resilience interventions might exert their effects. Resilience interventions have so far been developed to promote resilience by training putative resilience factors. However, the effects of these interventions in adult populations are small to medium and the evidence in younger populations is limited. Our Review shows that the conceptual and empirical foundations of these resilience interventions are misaligned with the leading theories in resilience research (regulatory flexibility^{18,138} and positive appraisal style theory^{19,25}), which focus on processes and outcomes rather than on factors. Integrating these leading theories with advances in transdiagnostic research and a developmental perspective might yield more durable intervention benefits, whereas dynamic modelling of how stressor types, intervention timing, cultural context and individual characteristics shape the outcomes of interventions might enable tailored approaches.

Moving forwards, a critical question is how resilience interventions should be developed to achieve meaningful public health impact. Our review of the implementation landscape reveals that most resilience interventions have not advanced to the later stages of the translational research pipeline¹¹⁷, thereby limiting their readiness for dissemination and implementation in real-world settings. More integrated and iterative research approaches might offer a promising path in resilience research¹¹⁴. Such approaches aim at overcoming the linearity between intervention development, efficacy and

effectiveness trials, and subsequent research efforts into implementation and dissemination¹¹⁷. Instead, they establish an open-ended process of interacting research phases²⁰³ that acknowledges participants' expertise rather than assuming an unidirectional transfer of knowledge from researchers to study participants. Future research should incorporate bottom-up intervention design and continuous evaluation in real-world settings²⁰⁴ — particularly when interventions pose minimal risk to participants.

For example, a low-threshold, messenger-based counselling service designed by young people in Germany to support peers experiencing heightened mental distress during the COVID-19 pandemic was found to be feasible and well accepted, with high user endorsement reflected in 85% of users reporting that they would recommend the service^{205,206}. Although these preliminary findings still require replication in large-scale effectiveness studies, they provide an example of how coping processes can inform and stimulate intervention development.

Learning from real-world coping processes is especially valuable in researching resilience mechanisms, which involves understanding outcomes in relation to acute exposure to novel and unforeseen stressors. Future research should therefore be guided by the co-creation of interventions with stressor-exposed individuals, key stakeholders and implementers (Box 3). It should also incorporate continuous feedback loops to adapt both interventions and implementation strategies.

Co-designed resilience interventions focused on mechanisms must be evaluated according to standards of evidence-based practice²⁰⁷, using trials with adequate active control groups accounting for unspecific intervention effects. Future evaluation studies should

Box 3 | Participatory approaches to developing interventions

Strengthening the real-world applicability and public-health impact of resilience interventions requires systematic testing and adaptation of the interventions in the contexts where they are intended to be used. Approaches such as community engagement, co-design with users^{222,223} and stakeholders, and living-lab or real-world experiments can support the development of interventions that fit local needs, are feasible to implement and are more likely to be sustained over time than interventions that are designed top-down by researchers. These participatory approaches also foster reciprocal capacity building within service systems, which can facilitate scale-up and long-term integration of the interventions into routine settings. Taken together, these approaches can enhance local ownership, improve scalability and increase the likelihood that the resilience interventions remain in use beyond the duration of the research programme.

Co-design and participatory development

Co-design approaches involve the active participation of users and stakeholders in identifying needs, shaping intervention components and refining implementation strategies²²². Such approaches enhance the contextual fit and relevance of the interventions, increasing acceptability and engagement.

For example, in school settings, a co-design process for a resilience intervention might involve students, teachers, social workers and school psychologists, all collaboratively determining how the intervention content can be integrated into the curriculum and adapted to the school climate and available resources.

Living labs and real-world experiments

Living labs provide structured environments for the iterative development, testing and refinement of interventions under naturalistic conditions^{224,225}. They enable cyclic improvement, multistakeholder engagement and on-site data collection — for example, using digital monitoring tools, organizational indicators or ecological assessments. These approaches enable researchers and implementers to jointly identify when intervention components require adaptation to local routines, constraints or stressor patterns.

For example, after a school-based resilience intervention has been co-designed with teachers, students and counsellors — who have collaboratively identified relevant stressors and develop feasible intervention components — a living-lab could extend this co-design approach by iteratively testing and refining these components within regular classroom routines based on continuous feedback, ultimately becoming embedded into school structures.

Reciprocal capacity building

Embedding resilience interventions within existing service systems requires reciprocal capacity building between researchers and organizations²²⁶. This process might involve training local staff to deliver or supervise intervention components, integrating activities into routine practices and establishing shared learning processes to evaluate feasibility and sustainability.

For example, paediatricians and paediatric nurses could integrate short resilience exercises into routine consultations for children and adolescents experiencing ongoing stress.

address key shortcomings in current intervention research by: precisely defining the target group; assessing and varying the timing of the intervention relative to stressor exposure; assessing implementation outcomes¹³⁰ along with effectiveness measures; systematically measuring stressor exposure before, during and after intervention delivery; and evaluating effects on real-world stressor responses as the primary outcome. Although such recommendations are not new in resilience research^{30,50,51}, they have yet to be implemented. To investigate the underlying change processes, ecological momentary data could be incorporated within large-scale trials. This method enables the assessment of individual-level network models, which can then be generalized to group-level findings using techniques such as group iterative multiple model estimation²⁰⁸. Such analyses hold the potential to transform our understanding of how resilience interventions work — ultimately supporting the design of more effective and scalable resilience interventions.

Individualized interventions might be used to tailor content and timing to stable cultural factors and individual characteristics or, through just-in-time adaptive interventions, to dynamic and situational mental states and ongoing stressor exposure^{156,209}. However, the limited understanding of the role of moderators on intervention effects hinders the development of such tailored strategies^{59,202}. The field could benefit from initiatives that create large-scale databases, where data from individual studies are catalogued and interoperably archived to enable their integration and facilitation of robust moderator analyses. Resilience researchers could take inspiration from collaborative initiatives in the field of transdiagnostic low-intensity psychosocial interventions for stressor-exposed populations²¹⁰. By specifically developing and examining resilience interventions in low- and middle-income countries as well as in marginalized groups in high-income contexts, where risk for stressor exposure is higher²¹¹, we could use resilience interventions to help to reduce health inequalities.

Finally, future research should place greater emphasis on a developmental perspective on resilience interventions. Our review of the literature, predominantly focused on adult samples, reflects the relative scarcity of intervention research in earlier developmental phases^{40,42,49}. However, resilience interventions must acknowledge developmental specificities and avoid a ‘one-size-fits-all’ approach. A multisystemic perspective is needed for advancing understanding of resilience mechanisms in children and adolescents, and for enhancing the effectiveness of interventions across the lifespan, from early childhood through older adulthood. Progress in this area will require a larger number of high-quality, long-term longitudinal observational studies across different developmental stages, assessing processes at the level of the individuals, their environments and their social contexts²¹². Such studies might focus on normative transitions occurring across the lifespan (such as school transitions or retirement) to assess pre-stressor baseline data and the subsequent processes²¹³. Although integrating developmental considerations might increase the complexity of study designs, it is crucial to unlocking the full potential of resilience interventions to prepare individuals — and, by extension, societies — to effectively cope with adversity.

Published online: 21 May 2026

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Acknowledgements

The authors thank R. Kalisch for valuable feedback on resilience mechanisms and S. Mrugalla for assistance in verifying the information presented in Tables 1 to 3 of this Review article.

Author contributions

S.K.S. led the writing and development of the core theses, with major contributions from J.S.-W. and K.L. All authors approved the final draft of this manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Peer review information *Nature Reviews Psychology* thanks Mark Shuquan Chen, Alexander C. McFarlane and Christian Rauschenberg for their contribution to the peer review of this work.

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